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Intelligent assistant content generation

Abstract

A computing system receives user activity data from a user interacting with content associated with a third party computing system. The computing system generates a profile for the user based on the user activity data. The profile includes tags indicating attributes about the user. The computing system establishes a communication session between the user and the intelligent assistant via a chat interface. The computing system receives a message from the user to the intelligent assistant. The computing system determines that the message includes at least one key word triggering retrieval of relevant content. Based on the determining, the computing system retrieves content that is associated with the key word and relevant to the user based on the tags associated with the user profile. The computing system causes a client device associated with the user to display the content alongside the chat interface.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/658,301, filed Apr. 7, 2022, which claims priority to U.S. Provisional Application Ser. No. 63/172,371, filed Apr. 8, 2021, which are hereby incorporated by reference in their entireties.

FIELD OF THE DISCLOSURE

(1) Embodiments disclosed herein generally related to a content generation system using an intelligent assistant.

BACKGROUND

(2) Today, content provided by third party systems is mostly static, requiring visitors to search for information. Likewise, when participating in an interactive chat with a human or automated chat, conversations are linear and scripted.

SUMMARY

(3) In some embodiments, a method is disclosed herein. A computing system receives user activity data from a user interacting with content associated with a third party computing system. The user activity is received via one or more integrations injected into code that is associated with an application presenting the content. The application includes a chat interface that links the user to an intelligent assistant hosted by the computing system. The computing system generates a profile for the user based on the user's activity data. The profile includes tags indicating attributes about the user. The computing system establishes a communication session between the user and the intelligent assistant via the chat interface. The computing system receives a message from the user to the intelligent assistant. The computing system determines that the message includes at least one key word triggering retrieval of relevant content. Based on the determining, the computing system retrieves content that is associated with the key word and relevant to the user based on the tags associated with the user's profile. The computing system causes a client device associated with the user to display the content alongside the chat interface.

(4) In some embodiments, a non-transitory computer readable medium is disclosed herein. The non-transitory computer readable medium includes instructions which, when executed by a computing system, cause a computing system to perform operations. The operations include receiving, from the computing system, activity data from a user interacting with content that is associated with a third party computing system. The user activity is received via one or more integrations injected into code that is associated with an application presenting the content. The application includes a chat interface linking the user to an intelligent assistant hosted by the computing system. The operations further include generating, by the computing system, a profile for the user based on the user's activity data. The profile includes tags indicating attributes about the user. The operations further include establishing, by the computing system, a communication session between the user and the intelligent assistant via the chat interface. The operations further include receiving, from the computing system, a message from the user to the intelligent assistant. The operations further include determining, by the computing system, that the message comprises at least one key word triggering retrieval of relevant content. The operations further include, based on the determining, retrieving, by the computing system, content that is associated with the key word and relevant to the user based on the tags associated with the user's profile. The operations further include causing, by the computing system, a client device associated with the user to display the content alongside the chat interface.

(5) In some embodiments, a system is disclosed herein. The system includes a processor and a memory. The memory includes programming instructions stored thereon, which, when executed by the processor, cause the system to perform operations. The operations include receiving activity data from a user interacting with content that is associated with a third party computing system. The user activity is received via one or more integrations injected into code that is associated with an application presenting the content. The application includes a chat interface linking the user to an intelligent assistant hosted by the computing system. The operations further include generating a profile for the user based on the user's activity data. The profile includes tags indicating attributes about the user. The operations further include establishing a communication session between the user and the intelligent assistant via the chat interface. The operations further include receiving a message from the user to the intelligent assistant. The operations further include determining that the message includes at least one key word triggering retrieval of relevant content. The operations further include, based on the determining, retrieving content that is associated with the key word and relevant to the user based on the tags associated with the user's profile. The operations further include causing a client device associated with the user to display the content alongside the chat interface.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) So that the manner in which the above recited features of the present disclosure can be understood in detail, a more particular description of the disclosure, briefly summarized above, may be had by reference to embodiments, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only typical embodiments of this disclosure and are therefore not to be considered limiting of its scope, for the disclosure may admit to other equally effective embodiments.

(2) FIG. 1 is a block diagram illustrating a computing environment, according to one exemplary embodiment.

(3) FIG. 2 is a block diagram illustrating the flow of information in computing environment **100**, according to example embodiments.

(4) FIG. 3A illustrates an example view of a graphical user interface presenting content hosted by a

third party system, according to example embodiments.

(5) FIG. 3B illustrates an example view of a graphical user interface presenting content hosted by a third party system, according to example embodiments.

(6) FIG. 3C illustrates an example view of a graphical user interface presenting content hosted by a third party system, according to example embodiments.

(7) FIG. 3D illustrates an example view of a graphical user interface presenting content hosted by a third party system, according to example embodiments.

(8) FIG. 3E illustrates an example view of a graphical user interface presenting content hosted by a third party system, according to example embodiments.

(9) FIG. 3F illustrates an example view of a graphical user interface presenting content hosted by a third party system, according to example embodiments.

(10) FIG. 4 is a flow diagram of a method of generating a graphical user interface, according to example embodiments.

(11) FIG. 5A illustrates a system bus computing system architecture, according to example embodiments.

(12) FIG. 5B illustrates a computer system having a chipset architecture, according to example embodiments.

(13) To facilitate understanding, identical reference numerals have been used, where possible, to designate identical elements that are common to the figures. It is contemplated that elements disclosed in one embodiment may be beneficially utilized on other embodiments without specific recitation.

DETAILED DESCRIPTION

(14) One or more techniques described herein improve upon conventional intelligent assistant experience systems by providing contextual information relevant to the user as it is discovered through a communication session with the intelligent assistant. For example, during the course of a communication session, the present system may provide the user with content relevant to topics discussed with the intelligent assistant. In this manner, the content displayed may be personalized for the user based on the data about such user.

(15) The term “user” as used herein includes, for example, a person or entity that owns a computing device or wireless device; a person or entity that operates or utilizes a computing device or wireless device; or a person or entity that is otherwise associated with a computing device or wireless device. It is contemplated that the term “user” is not intended to be limiting and may include various examples beyond those described.

(16) FIG. 1 is a block diagram illustrating computing environment **100**, according to one embodiment. Computing environment **100** may include at least one or more client devices **102**, an organization computing system **104**, and a third party system **106** communicating via network **105**.

(17) Network **105** may be of any suitable type, including individual connections via the Internet, such as cellular or Wi-Fi networks. In some embodiments, network **105** may connect terminals, services, and mobile devices using direct connections, such as radio frequency identification (RFID), near-field communication (NFC), Bluetooth™, low-energy Bluetooth™ (BLE), Wi-Fi™, ZigBee™, ambient backscatter communication (ABC) protocols, USB, WAN, or LAN. Because the information transmitted may be personal or confidential, security concerns may dictate one or more of these types of connection be encrypted or otherwise secured. In some embodiments, however, the information being transmitted may be less personal, and therefore, the network connections may be selected for convenience over security.

(18) Network **105** may include any type of computer networking arrangement used to exchange data. For example, network **105** may be the Internet, a private data network, virtual private network using a public network and/or other suitable connection(s) that enables components in computing environment **100** to send and receive information between the components of computing environment **100**.

(19) Client device **102** may be operated by a user. For example, client device **102** may be a mobile device, a tablet, a desktop computer, or any computing system having the capabilities described herein. Client device **102** may belong to or be provided to a customer or may be borrowed, rented, or shared. Customers may include individuals such as, for example, subscribers, clients, prospective clients, or prospective job candidates of an entity associated with third party system **106**, such as individuals who may be interested in employment with an entity associated with third party systems **106**.

(20) Client device **102** may include at least application **110** and web browser **112**. In some embodiments, application **110** may be a standalone application associated with third party system **106**. In some embodiments, web browser **112** may allow access to a website associated with third party system **106**. Client device **102** may access application **110** or web browser **112** to access content associated with third party system **106**. In some embodiments, client device **102** may communicate over network **105** to request a webpage, for example, from web client application server **114** of third party system **106**. For example, client device **102** may be configured to execute application **110** or web browser **112** to access content managed by web client application server **114**. The content that is displayed to client device **102** may be transmitted from web client application server **114** to client device **102**, and subsequently processed by application **110** or web browser **112** for display through a graphical user interface (GUI) of client device **102**.

(21) Third party system **106** may include at least web client application server **114**, intelligent assistant integration **116**, and customization integration **118**. Intelligent assistant integration **116** may allow third party system **106** to incorporate an intelligent assistant associated with organization computing system **104** into a website associated with third party system **106** or content presented via application **110** associated with third party system **106**. For example, an intelligent assistant may be incorporated into webpages of websites associated with third party system **106**, web-based platforms, messaging applications (e.g., Facebook Messenger, WhatsApp®, Signal, and the like), a mobile application (e.g., standalone application **110**), a short message service (SMS) application, a multimedia messaging service (MMS) application, and the like. For example, intelligent assistant integration **116** may take the form of code that may be injected into the web code of a website hosted by web client application server **114**. In another example, intelligent assistant integration **116** may take the form of code that may be injected into the code of application **110** executing on client device **102**. Once injected into the web code of the website or code of application **110**, an intelligent assistant supported by organization computing system **104** may be incorporated into content provided by third party system **106**. As such, when a user of client device **102** accesses application **110**, application **110** may render an intelligent assistant associated with organization computing system **104**. In some embodiments, when a user of client device **102** requests a website from web client application server **114**, web browser **112** may render an intelligent assistant associated with organization computing system **104** within the website.

(22) Customization integration **118** may allow third party system **106** to provide customized content to a user interacting with a website or application **110** associated with third party system **106**. For example, customization integration **118** may take the form of code that is injected to the web code of a website or code of application **110** associated with third party system **106**. In some embodiments, once injected into the web code of the website, when the user of client device **102** requests the web page from web client application server **114**, customization integration **118** may allow organization computing system **104** to track user activity on the website. In some embodiments, once injected into the code of application **110**, when user of client device **102** accesses application **110**, customization integration **118** may allow organization computing system **104** to track user activity while interacting with application **110**. For example, customization integration **118** may allow organization computing system **104** to track document upload (e.g., resume upload), internet protocol (IP) information (e.g., device location), campaigns (e.g., sites the user visited from), pages of the website viewed, portions of the website viewed (e.g., job postings),

time spent on webpages of a website, browser language, and the like. In this manner, customization integration **118** may enable organization computing system **104** to provide users with curated content tailored to the users' needs.

(23) Organization computing system **104** may include at least web client application server **124**, intelligent assistant **126**, customization module **128**, and natural language processing (NLP) module **130**. Each of intelligent assistant **126**, customization module **128**, and NLP module **130** may be comprised of one or more software modules. The one or more software modules may be collections of code or instructions stored on a media (e.g., memory of organization computing system **104**) that represent a series of machine instructions (e.g., program code) that implements one or more algorithmic steps. Such machine instructions may be the actual computer code the processor of organization computing system **104** interprets to implement the instructions or, alternatively, may be a higher level of coding of the instructions that is interpreted to obtain the actual computer code. The one or more software modules may also include one or more hardware components. One or more aspects of an example algorithm may be performed by the hardware components (e.g., circuitry) itself, rather than as a result of instructions.

(24) In some embodiments, when client device **102** requests a website from web client application server **114** from third party system **106**, web client application server **114** may provide web browser **112** of client device **102** with web code associated with the web page. For example, web browser **112** may transmit a hypertext transfer protocol (HTTP) request to web client application server **114**. Web client application server **114** may read the request and transmit an HTTP message back to application **110**. The response from web client application server **114** may include hypertext text markup language (HTML) code corresponding to the website. The HTML code may include the web code associated with intelligent assistant integration **116** and customization integration **118**. Accordingly, when web browser **112** processes the HTML code for presentation of the web site to the user, a connection may be established between client device **102** and organization computing system **104**.

(25) In some embodiments, when client device **102** accesses content associated with third party system **106** via application **110**, application **110** may utilize one or more application programming interfaces (APIs) to access functionality of intelligent assistant integration **116** and customization integration **118**. Accordingly, when application **110** establishes a connection with back end computing system **104** via one or more APIs, a connection may be established between client device **102** and back end computing system **104**.

(26) Intelligent assistant **126** may be configured to interact with a user while the user browses the webpage provided by third party system **106**. For example, as a user interacts with an interface associated with intelligent assistant **126**, intelligent assistant **126** may receive messages from client device **102**. Intelligent assistant **126** may utilize one or more natural language processing and machine learning techniques to determining a meaning of the user's message and generate an appropriate message for response. In some embodiments, intelligent assistant **126** may be configured to process the messages provided by the user to identify one or more keywords contained in the message. The one or more keywords identified by intelligent assistant **126** may be utilized by customization module **128** to provide customized content to the user.

(27) Customization module **128** may be configured to retrieve customized content for the user. In some embodiments, customization module **128** may retrieve customized content for the user based on one or more of keywords and/or phrases provided by the user to intelligent assistant **126**, as well as activity streams provided by customization integration **118** to customization module **128**. For example, the activity stream may include, but is not limited to, document upload (e.g., resume upload), internet protocol (IP) information (e.g., device location), campaigns (e.g., websites the user visited from), pages of the website viewed, portions of the website viewed (e.g., job postings), time spent on each webpage of the website, browser language, cursor movements, portions of the application viewed, time spent on each portion of the application, and the like.

(28) Customization module **128** may be configured to build a profile for the user. In some embodiments, customization module **128** may build a profile for the user in real-time (or near real-time) based on the one or more keywords and/or phrases and the activity using one or more heuristics to identify content to be provided to the user. The profile may include attributes about the user, such as, but not limited to, location information, employment history, age, gender, educational background, areas of interest, and the like. In some embodiments, the profile may further include one or more tags corresponding to attributes of the user. The one or more tags may allow customization module **128** to retrieve relevant content from database **108** for presentation to the user.

(29) In some embodiments, the content provided to user of client device **102**, either via application **110** or web browser **112**, may include an option for users to upload documents. Exemplary documents may include, but are not limited to, resumes, cover letters, school transcripts, recommendation letters, completed questionnaires, and the like. NLP module **130** may be configured to analyze and process documents uploaded by an end user via the website. In some embodiments, the user may upload a document when prompted by intelligent assistant **126**. NLP module **130** may be configured to extract information that may be used by customization module **128** for generating the profile of the user. For example, NLP module **130** may extract information directed to: employment history, education history, job skills, keywords, and the like. NLP module **130** may provide the extracted information to customization module **128** for continued generation of the user's profile.

(30) In some embodiments, customization module **128** may be in communication with database **108**. Database **108** may be configured to store content associated with various clients **140**. Each client **140** may correspond to an entity associated with a respective third party system **106**. In this manner, each third party system **106** may store content to be delivered to end users in database **108** associated with organization computing system **104**, such that customization module **128** may retrieve relevant information to provide to an end user.

(31) As shown, each client **140** may include content **142** and users **144**. Content **142** may correspond to content unique to a client associated with a respective third party system **106**. In some embodiments, content **142** may include information unique to the client. Using a specific example, content **142** may include information unique to the onboarding or hiring process of the client, such as, but not limited to, human resource information. Content **142** may include one or more tags **146**. Tags **146** may correspond to attributes or categories to be associated with a respective piece of content **142**. In this manner, customization module **128** may match one or more tags associated with the user's profile to tags **146** of content **142**. Tags **146** may allow customization module **128** to retrieve relevant content from database **108** for presentation to the user.

(32) Users **144** may correspond to one or more users that have accessed content associated with third party system **106**. In some embodiments, users **144** may be representative of one or more user profiles generated by customization module **128**. Each user **144** may include one or more tags **148** associated therewith. As provided above, tags **148** may allow customization module **128** to retrieve relevant content **142** from database **108** for presentation to the user. For example, when providing relevant content to a user, customization module **128** may match tags **148** associated with the user with tags **146** associated with content **142**. In this manner, customization module **128** may provide the user with relevant content, given the context of the user's interaction with intelligent assistant **126**.

(33) FIG. 2 is a block diagram illustrating flow **200** of information in computing environment **100**, according to example embodiments. As shown, organization computing system **104** may receive, as inputs, user activity **202**, intelligent assistant activity **204**, and document uploads **206**. User activity **202** may correspond to one or more of IP information (e.g., device location), campaigns (e.g., websites the user visited from), pages of the website viewed, portions of the website viewed (e.g.,

job postings), time spent on each webpage of the website, browser language, cursor movements, portions of content viewed via application **110**, time spent on each portion of content in application **110**, and the like. User activity **202** may be received at organization computing system **104** using customization integration **118**.

(34) Intelligent assistant activity **204** may correspond to communications between an end user of client device **102** and intelligent assistant **126**. As a user interacts with an interface associated with intelligent assistant **126**, intelligent assistant **126** may receive messages from client device **102**. Intelligent assistant **126** may utilize one or more natural language processing and machine learning techniques to determine a meaning of the user's message and generate an appropriate message for response. In some embodiments, intelligent assistant **126** may be configured to process the messages provided by the user to identify one or more keywords contained in the message. The one or more keywords identified by intelligent assistant **126** may be utilized by customization module **128** for providing customized content to the user.

(35) Document uploads **206** may correspond to one or more documents uploaded by a user during the user's communication session with intelligent assistant **126**. Documents may include, but are not limited to, resumes, cover letters, school transcripts, recommendation letters, completed questionnaires, and the like. NLP module **130** may receive and analyze document uploads **206**. NLP module **130** may extract information that may be used by customization module **128** for generating the profile of the user. For example, NLP module **130** may extract information directed to: employment history, education history, job skills, keywords, and the like. NLP module **130** may provide the extracted information to customization module **128** for continued generation of the user's profile.

(36) Importantly, customization module **128** does not need to wait until it receives all information from intelligent assistant **126**, NLP module **130**, or customization integration **118**. Instead, customization module **128** may be configured to continuously and dynamically generate and update a profile of the user based on incoming streams of user activity data. Accordingly, customization module **128** may continuously update the user's profile and retrieve relevant information from database **108** for presentation to the user based on the continuously updated profile.

(37) In some embodiments, customization module **128** may retrieve content from database **108** based on the current context of a dialogue established between intelligent assistant **126** and client device **102**. For example, customization module **128** may retrieve certain content from database **108**, based on one or more key words or phrases identified by intelligent assistant **126**. In this manner, customization module **128** may serve the user with content relevant to the user's conversation with intelligent assistant **126**, in real-time, as the conversation progresses.

(38) FIG. 3A illustrates an example view of a graphical user interface **300** (hereinafter “GUI **300**”) presenting content hosted by third party system **106**, according to example embodiments. In some embodiments, GUI **300** may be a webpage presented in web browser **112** of client device **102**. In some embodiments, GUI **300** may be a graphical user interface generated by application **110** executing on client device **102**.

(39) As illustrated, GUI **300** may present content **302** (e.g., a webpage or portions of an application) to the user. As shown, content **302** may correspond to a hiring page associated with third party system **106**. GUI **300** may further include an intelligent assistant prompt **304**. Intelligent assistant prompt **304** may be representative of a graphical element, which, when interacted with, may establish an interactive communication session with intelligent assistant **126**.

(40) FIG. 3B illustrates an example view of a graphical user interface **320** presenting content hosted by third party system **106**, according to example embodiments. Graphical user interface **320** (hereinafter “GUI **320**”) may correspond to an updated view responsive to the user interacting with intelligent assistant prompt **304** in GUI **300**. In some embodiments, GUI **320** may be a webpage presented in web browser **112** of client device **102**. In some embodiments, GUI **320** may be a graphical user interface generated by application **110** executing on client device **102**.

(41) As illustrated, GUI **320** may include an intelligent assistant area **322**. Intelligent assistant area **322** may be representative of a chat interface or chat integration that links the user to functionality of intelligent assistant **126**. Intelligent assistant **126** may initiate the conversation with the user by sending one or more messages **324** to the user. One or more messages **324** may be used to gather additional details from the user during the conversation.

(42) FIG. **3C** illustrates an example view of a graphical user interface **330** presenting content hosted by third party system **106**, according to example embodiments. Graphical user interface **330** (hereinafter “GUI **330**”) may correspond to an updated view responsive to the intelligent assistant **126** providing one or more messages **324** to client device **102**. In some embodiments, GUI **330** may be a webpage presented in web browser **112** of client device **102**. In some embodiments, GUI **330** may be a graphical user interface generated by software application **110** executing on client device **102**.

(43) As shown, a user may respond to one or more messages **324** with one or more messages **332**. One or more messages **332** may be provided to intelligent assistant **126** using the chat interface of intelligent assistant area **322**. One or more messages **332** may be provided to intelligent assistant **126** for further analysis.

(44) FIG. **3D** illustrates an example view of a graphical user interface **340** presenting content hosted by third party system **106**, according to example embodiments. Graphical user interface **340** (hereinafter “GUI **340**”) may correspond to an updated view responsive to customization module **128** providing customized content to client device **102**. In some embodiments, GUI **340** may be a webpage presented in web browser **112** of client device **102**. In some embodiments, GUI **340** may be a graphical user interface generated by application **110** executing on client device **102**.

(45) As shown, responsive to intelligent assistant **126** receiving message **332** from client device **102**, intelligent assistant **126** may process message **332** to identify one or more key words or trigger words contained therein. Message **332** recites: “Hi Olivia, I’m looking for a job.” The word “job” may be associated with a key word or trigger word that may prompt intelligent assistant **126** to instruct customization module **128** to provide customized content to client device **102**. Based on identifying the key word or trigger word, customization module **128** may identify relevant content for the user from database **108**. Customization module **128** may provide the relevant content to client device **102** for display via web browser **112** or application **110**. For example, as shown, relevant content **346** may be displayed adjacent to intelligent assistant area **322**. In this manner, a user may be provided with content relevant to their conversation with intelligent assistant **126**.

(46) In some embodiments, intelligent assistant **126** may further prompt a user of client device **102** for further information. For example, as shown, intelligent assistant **126** may prompt a user to upload relevant documents. To do so, a document upload prompt **344** may appear within the webpage. As illustrated, document upload prompt **344** may appear within intelligent assistant area **322**. In this manner, a user may drag and drop relevant documents within intelligent assistant area **322**. Organization computing system **104** may review and analyze the relevant documents to determine whether any additional content may be provided to the user.

(47) FIG. **3E** illustrates an example view of a graphical user interface **350** presenting content hosted by third party system **106**, according to example embodiments. Graphical user interface **350** (hereinafter “GUI **350**”) may correspond to an updated view responsive to a user uploading a document. In some embodiments, GUI **350** may be a webpage presented in web browser **112** of client device **102**. In some embodiments, GUI **350** may be a graphical user interface generated by application **110** executing on client device **102**.

(48) Responsive to receiving a document via document upload prompt **344**, NLP module **130** may review and analyze the contents of the document. NLP module **130** may provide information about the document to customization module **128** for retrieval of relevant information. For example, based on analyzing the document, NLP module **130** may determine that the user has experience with product management. As such, NLP module **130** may prompt customization module **128** to

retrieve content related to product management jobs.

(49) As shown, customization module **128** may retrieve content relevant to product management positions to present within the webpage. In some embodiments, customization module **128** may cause client device **102** to display the product management position information within intelligent assistant area **322** (as shown by reference **354**). In some embodiments, customization module **128** may cause client device **102** to display the product management position information adjacent to intelligent assistant area **322** (as shown by “**356**” in FIG. **3E**).

(50) In some embodiments, NLP module **130** may further prompt intelligent assistant **126** with respect to the contents of the document. For example, as shown, responsive to NLP module **130** determining that the user has experience with product management, intelligent assistant **126** may send the user message **352**, further engaging the user based on the newly discovered information.

(51) In some embodiments, new content may be provided, and indicator **358** may be added the displayed content. Indicator **358** may correspond to content previously presented within the web browser **112** or application **110**. For example, indicator **358** may correspond to the Welcome content that is shown in GUI **340**. As such, when a user interacts with indicator **358**, GUI **350** may be updated to present content **346**, i.e., content corresponding to indicator **358**.

(52) FIG. **3F** illustrates an example view of a graphical user interface **360** presenting content hosted by third party system **106**, according to example embodiments. Graphical user interface **360** (hereinafter “GUI **360**”) may correspond to an updated view responsive to a user submitting a question to intelligent assistant **126**. In some embodiments, GUI **360** may be a webpage presented in web browser **112** of client device **102**. In some embodiments, GUI **360** may be a graphical user interface generated by application **110** executing on client device **102**.

(53) As shown, a user may generate a message **362** for intelligent assistant **126**. Message **362** may seek further information about the hiring process. Responsive to receiving message **362**, intelligent assistant **126** may respond with message **364** to be presented to the user. Additionally, customization module **128** may be prompted to serve additional content **368** to the user. Content **368** may be relevant to the question asked by the user. In some embodiments, customization module **128** may be prompted to retrieve additional content, responsive to intelligent assistant **126** identifying a key word or trigger word in message **362**. In the example shown in FIG. **3F**, the user is seeking additional information about the hiring process.

(54) As shown, content **368** may be reflective of information about the hiring process. Content **368** may take the form of one or more images, documents, videos, or audio files that may be presented to the user within the webpage.

(55) At any point, the user may respond with follow up messages, such as that illustrated as message **366**.

(56) Importantly, as the user is interacting with intelligent assistant **126** and content presented to the user, customization module **128** may be continually building or updating a profile associated with the user. As customization module **128** learns additional information about the user, customization module **128** may be able to provide the user with more targeted content.

(57) In some embodiments, as new content is provided, additional indicators (e.g., indicator **370**) may be added the displayed content. Indicator **370** may correspond to content previously presented within the web browser **112** or application **110**. For example, indicator **370** may correspond to the job content that is shown in GUI **350**. As such, when a user interacts with indicator **370**, GUI **360** may be updated to present content **356**, i.e., content corresponding to indicator **370**. Additionally, as shown, indicator **370** is added to GUI **360**, adjacent the previously generated indicator **358**. In this manner, a user can easily navigate between content.

(58) FIG. **4** is a flow diagram illustrating a method **400** of generating an interactive graphical user interface (GUI) according to example embodiments. Method **400** may begin at step **402**.

(59) At step **402**, organization computing system **104** may receive user data from a user interacting with content hosted by third party system **106**. In some embodiments, user data may be

representative of user activity, intelligent assistant activity, and document uploads.

(60) User activity may include one or more of IP information (e.g., device location), campaigns (e.g., sites the user visited from), pages of the website viewed, portions of the website viewed (e.g., job postings), time spent on each webpage of the website, browser language, cursor movements, portions of the application viewed, time spent on each portion of the application, and the like. User activity may be received at organization computing system **104** using customization integration **118**.

(61) Intelligent assistant activity **204** may include communications between an end user of client device **102** and intelligent assistant **126**. For example, intelligent assistant activity **204** may include messages received from client device **102**.

(62) Document uploads **206** may include one or more documents uploaded by a user during the user's communication session with intelligent assistant **126**. Exemplary documents may include, but are not limited to, resumes, cover letters, school transcripts, recommendation letters, completed questionnaires, and the like.

(63) At step **404**, organization computing system **104** may process the user data received. In some embodiments, customization module **128** may log the user activity to identify potential interests of the user. In some embodiments, intelligent assistant **126** may utilize one or more natural language processing and machine learning techniques to determine a meaning of the user's message and generate an appropriate message for response. In some embodiments, intelligent assistant **126** may be configured to process the messages provided by the user to identify one or more keywords contained in the message. The one or more keywords identified by intelligent assistant **126** may be utilized by customization module **128** for providing customized content to the user.

(64) In some embodiments, NLP module **130** may receive and analyze document uploads **206**. NLP module **130** may extract information that may be used by customization module **128** for generating the profile of the user. For example, NLP module **130** may extract information directed to: employment history, education history, job skills, keywords, and the like. NLP module **130** may provide the extracted information to customization module **128** for continued generation of the user's profile.

(65) At step **406**, organization computing system **104** may generate or update a profile for the user based on the user's activity data. In some embodiments, customization module **128** may build a profile for the user in real-time (or near real-time) based on the one or more keywords and/or phrases and the activity, using one or more heuristics to identify content to be provided to the user. The profile may include attributes about the user, such as, but not limited to location information, employment history, age, gender, educational background, areas of interest, and the like. In some embodiments, the profile may further include one or more tags corresponding to attributes of the user. The one or more tags may allow customization module **128** to retrieve relevant content from database **108** for presentation to the user.

(66) At step **408**, organization computing system **104** may determine whether a trigger is identified. In some embodiments, a trigger may correspond to a key word or trigger word in a message sent by client device **102** to intelligent assistant **126**. If, at step **408**, customization module **128** determines that a trigger is not present, then method **400** may proceed to step **410** and organization computing system **104** may wait for additional user activity data.

(67) If, however, at step **408**, customization module **128** determines that a trigger is present, i.e., a user's message contains a key word or trigger word, then method **400** may proceed to step **412**. At step **412**, organization computing system **104** may retrieve relevant content for presentation to the user. For example, customization module **128** may identify the type of content suitable for the user based on the key word or trigger word. Customization module **128** may further tailor the selection of relevant content based on the generated profile of the user. For example, if the key word or trigger word is "looking for a job," customization module **128** may identify content related to job openings. Customization module **128** may select the specific content based on the profile of the

user. For example, customization module **128** may determine, based on the user's uploading of a resume, that the user has experience in product management. Accordingly, customization module **128** may generate a tag in the user's profile corresponding to product management. As such, when customization module **128** retrieves the job information from database **108**, customization module **128** may search for job information content relevant to product management based on the tag.

(68) At step **416**, organization computing system **104** may cause the relevant content to be displayed. For example, customization module **128** may provide web browser **112** of client device **102** with the relevant content for display in a webpage currently being viewed by the user. In another example, customization module **128** may provide application **110** of client device **102** with the relevant content for display with content presented via application **110**.

(69) At step **418**, organization computing system **104** may determine whether the dialogue ended. In other words, organization computing system **104** may determine whether the user has stopped interacting with intelligent assistant **126** or has otherwise terminated the conversation. If, at step **418**, organization computing system **104** determines that the dialogue has not ended, then method **400** may proceed at step **410** and organization computing system **104** may wait for additional user activity data. If, however, at step **418**, organization computing system **104** determines that the dialogue has ended, then method **400** may end.

(70) FIG. 5A illustrates an architecture of system bus computing system **500**, according to example embodiments. One or more components of system **500** may be in electrical communication with each other using a bus **505**. System **500** may include a processor (e.g., one or more CPUs, GPUs or other types of processors) **510** and a system bus **505** that couples various system components including the system memory **515**, such as read only memory (ROM) **520** and random access memory (RAM) **525**, to processor **510**. System **500** can include a cache of high-speed memory connected directly with, in close proximity to, or integrated as part of processor **510**. System **500** can copy data from memory **515** and/or storage device **530** to cache **512** for quick access by processor **510**. In this way, cache **512** may provide a performance boost that avoids processor **510** delays while waiting for data. These and other modules can control or be configured to control processor **510** to perform various actions. Other system memory **515** may be available for use as well. Memory **515** may include multiple different types of memory with different performance characteristics. Processor **510** may be representative of a single processor or multiple processors. Processor **510** can include one or more of a general purpose processor or a hardware module or software module, such as service 1 **532**, service 2 **534**, and service 5 **536** stored in storage device **530**, configured to control processor **510**, as well as a special-purpose processor where software instructions are incorporated into the actual processor design. Processor **510** may essentially be a completely self-contained computing system, containing multiple cores or processors, a bus, memory controller, cache, etc. A multi-core processor may be symmetric or asymmetric.

(71) To enable user interaction with the system **500**, an input device **545** can be any number of input mechanisms, such as a microphone for speech, a touch-sensitive screen for gesture or graphical input, keyboard, mouse, motion input, speech and so forth. An output device **535** (e.g., a display) can also be one or more of a number of output mechanisms known to those of skill in the art. In some instances, multimodal systems can enable a user to provide multiple types of input to communicate with system **500**. Communication interface **540** can generally govern and manage the user input and system output. There is no restriction on operating on any particular hardware arrangement and therefore the basic features here may easily be substituted for improved hardware or firmware arrangements as they are developed.

(72) Storage device **530** may be a non-volatile memory and can be a hard disk or other type of computer readable media that can store data that are accessible by a computer, such as magnetic cassettes, flash memory cards, solid state memory devices, digital versatile disks, cartridges, random access memories (RAMs) **525**, read only memory (ROM) **520**, and hybrids thereof.

(73) Storage device **530** can include services **532**, **534**, and **536** for controlling the processor **510**.

Other hardware or software modules are contemplated. Storage device **530** can be connected to system bus **505**. In one aspect, a hardware module that performs a particular function can include the software component stored in a computer-readable medium in connection with the necessary hardware components, such as processor **510**, bus **505**, output device **535** (e.g., a display), and so forth, to carry out the function.

(74) FIG. **5B** illustrates a computer system **550** having a chipset architecture, according to example embodiments. Computer system **550** may be an example of computer hardware, software, and firmware that can be used to implement the disclosed technology. System **550** can include one or more processors **555**, representative of any number of physically and/or logically distinct resources capable of executing software, firmware, and hardware configured to perform identified computations. One or more processors **555** can communicate with a chipset **560** that can control input to and output from one or more processors **555**. In this example, chipset **560** outputs information to output **565**, such as a display, and can read and write information to storage device **570**, which can include magnetic media, and solid-state media, for example. Chipset **560** can also read data from and write data to storage device **575** (e.g., RAM). A bridge **580** for interfacing with a variety of user interface components **585** can be provided for interfacing with chipset **560**. Such user interface components **585** can include a keyboard, a microphone, touch detection and processing circuitry, a pointing device, such as a mouse, and so on. In general, inputs to system **550** can come from any of a variety of sources, machine generated and/or human generated.

(75) Chipset **560** can also interface with one or more communication interfaces **590** that can have different physical interfaces. Such communication interfaces can include interfaces for wired and wireless local area networks, for broadband wireless networks, as well as personal area networks. Some applications of the methods for generating, displaying, and using the GUI disclosed herein can include receiving ordered datasets over the physical interface or be generated by the machine itself by one or more processors **555** analyzing data stored in storage device **570** or **575**. Further, the machine can receive inputs from a user through user interface components **585** and execute appropriate functions, such as browsing functions by interpreting these inputs using one or more processors **555**.

(76) It can be appreciated that example systems **500** and **550** can have more than one processor **510** or be part of a group or cluster of computing devices networked together to provide greater processing capability.

(77) While the foregoing is directed to embodiments described herein, other and further embodiments may be devised without departing from the basic scope thereof. For example, aspects of the present disclosure may be implemented in hardware or software or a combination of hardware and software. One embodiment described herein may be implemented as a program product for use with a computer system. The program(s) of the program product define functions of the embodiments (including the methods described herein) and can be contained on a variety of computer-readable storage media. Illustrative computer-readable storage media include, but are not limited to: (i) non-writable storage media (e.g., read-only memory (ROM) devices within a computer, such as CD-ROM disks readably by a CD-ROM drive, flash memory, ROM chips, or any type of solid-state non-volatile memory) on which information is permanently stored; and (ii) writable storage media (e.g., floppy disks within a diskette drive or hard-disk drive or any type of solid state random-access memory) on which alterable information is stored. Such computer-readable storage media, when carrying computer-readable instructions that direct the functions of the disclosed embodiments, are embodiments of the present disclosure.

(78) It will be appreciated to those skilled in the art that the preceding examples are exemplary and not limiting. It is intended that all permutations, enhancements, equivalents, and improvements thereto are apparent to those skilled in the art upon a reading of the specification and a study of the drawings are included within the true spirit and scope of the present disclosure. It is therefore

intended that the following appended claims include all such modifications, permutations, and equivalents as fall within the true spirit and scope of these teachings.

Claims

1. A method, comprising: receiving, by a computing system, user activity data from a user of a user device interacting with content hosted by a third-party computing system via one or more integrations injected into code of an application executing on the user device, the application presenting the content to the user; learning, by the computing system, attributes of the user based on the user activity data; causing, by the computing system, the application to render a web page for display to the user via the user device, the web page comprising a chat interface at least partially overlaying the content of the web page; establishing, by the computing system, a communication session between the user and an intelligent assistant associated with the computing system via a chat interface of the application; continuing, by the computing system, to learn additional attributes of the user based on updated user activity data received via messages sent by the user to the intelligent assistant via the communication session between the user and the intelligent assistant and one or more of user browsing activity on the web page or location information derived from communications with the user; receiving, by the computing system, a message from the user to the intelligent assistant; and dynamically causing, by the computing system, the application to render customized content in the web page for the user based on the learned attributes and the learned additional attributes of the user by: determining, by the computing system, that the message comprises a key word triggering retrieval of content relevant to the message, based on the determining, retrieving, by the computing system, the relevant content that is associated with the key word and relevant to the user based on the learned attributes and the learned additional attributes of the user, and causing, by the computing system, the application to render customized content in the web page, the customized content comprising the content determined to be relevant to the user based on the message alongside the chat interface.
2. The method of claim 1, wherein the user activity data comprises at least one of: internet protocol (IP) information, campaign data, pages of a website viewed, time spent on each webpage, browser language, cursor movements, portions of the application viewed, or time spent on each portion of the application.
3. The method of claim 1, wherein learning, by the computing system, the attributes of the user further comprises: generating a plurality of tags corresponding to the learned attributes.
4. The method of claim 3, wherein retrieving, by the computing system, the relevant content is further based on a match between the plurality of tags and content tags associated with the relevant content.
5. The method of claim 1, wherein the communication session is established in response to the user interacting with an intelligent assistant prompt displayed within the content hosted by the third-party computing system.
6. The method of claim 1, wherein determining, by the computing system, that the message comprises the key word triggering retrieval of the relevant content comprises: analyzing, by a natural language processing module of the computing system, the message from the user to identify the key word.
7. The method of claim 1, further comprising: prompting, by the intelligent assistant, the user to upload a document, and wherein the learned attributes of the user is further based on analysis of the uploaded document.
8. A non-transitory computer readable medium comprising one or more sequences of instructions, which, when executed by a processor, causes a computing system to perform operations comprising: receiving, by the computing system, user activity data from a user of a user device interacting with content hosted by a third-party computing system via one or more integrations

injected into code of an application executing on the user device, the application presenting the content to the user; learning, by the computing system, attributes of the user based on the user activity data; causing, by the computing system, the application to render a web page for display to the user via the user device, the web page comprising a chat interface at least partially overlaying the content of the web page; establishing, by the computing system, a communication session between the user and an intelligent assistant associated with the computing system via a chat interface of the application; continuing, by the computing system, to learn additional attributes of the user based on updated user activity data received via messages sent by the user to the intelligent assistant via the communication session between the user and the intelligent assistant and one or more of user browsing activity on the web page or location information derived from communications with the user; receiving, by the computing system, a message from the user to the intelligent assistant; and dynamically causing, by the computing system, the application to render customized content in the web page for the user based on the learned attributes and the learned additional attributes of the user by: determining, by the computing system, that the message comprises a key word triggering retrieval of content relevant to the message, based on the determining, retrieving, by the computing system, the relevant content that is associated with the key word and relevant to the user based on the learned attributes and the learned additional attributes of the user, and causing, by the computing system, the application to render customized content in the web page, the customized content comprising the content determined to be relevant to the user based on the message alongside the chat interface.

9. The non-transitory computer readable medium of claim 8, wherein the user activity data comprises at least one of: internet protocol (IP) information, campaign data, pages of a website viewed, time spent on each webpage, browser language, cursor movements, portions of the application viewed, or time spent on each portion of the application.

10. The non-transitory computer readable medium of claim 8, wherein learning, by the computing system, the attributes of the user further comprises: generating a plurality of tags corresponding to the learned attributes.

11. The non-transitory computer readable medium of claim 10, wherein retrieving, by the computing system, the relevant content is further based on a match between the plurality of tags and content tags associated with the relevant content.

12. The non-transitory computer readable medium of claim 8, wherein the communication session is established in response to the user interacting with an intelligent assistant prompt displayed within the content hosted by the third-party computing system.

13. The non-transitory computer readable medium of claim 8, wherein determining, by the computing system, that the message comprises the key word triggering retrieval of the relevant content comprises: analyzing, by a natural language processing module of the computing system, the message from the user to identify the key word.

14. The non-transitory computer readable medium of claim 8, further comprising: prompting, by the intelligent assistant, the user to upload a document, and wherein the learned attributes of the user is further based on analysis of the uploaded document.

15. A system, comprising: a processor; and a memory having programming instructions stored thereon, which, when executed by the processor, causes the system to perform operations comprising: receiving user activity data from a user of a user device interacting with content hosted by a third-party computing system via one or more integrations injected into code of an application executing on the user device, the application presenting the content to the user; learning attributes of the user based on the user activity data; causing the application to render a web page for display to the user via the user device, the web page comprising a chat interface at least partially overlaying the content of the web page; establishing a communication session between the user and an intelligent assistant associated with the system via a chat interface of the application; continuing to learn additional attributes of the user based on updated user activity data received via messages

sent by the user to the intelligent assistant via the communication session between the user and the intelligent assistant and one or more of user browsing activity on the web page or location information derived from communications with the user; receiving a message from the user to the intelligent assistant; and dynamically causing the application to render customized content in the web page for the user based on the learned attributes and the learned additional attributes of the user by: determining that the message comprises a key word triggering retrieval of content relevant to the message, based on the determining, retrieving the relevant content that is associated with the key word and relevant to the user based on the learned attributes and the learned additional attributes of the user, and causing the application to render customized content in the web page, the customized content comprising the content determined to be relevant to the user based on the message alongside the chat interface.

16. The system of claim 15, wherein the user activity data comprises at least one of: internet protocol (IP) information, campaign data, pages of a website viewed, time spent on each webpage, browser language, cursor movements, portions of the application viewed, or time spent on each portion of the application.

17. The system of claim 15, wherein learning the attributes of the user further comprises: generating a plurality of tags corresponding to the learned attributes.

18. The system of claim 17, wherein retrieving the relevant content is further based on a match between the plurality of tags and content tags associated with the relevant content.

19. The system of claim 15, wherein the communication session is established in response to the user interacting with an intelligent assistant prompt displayed within the content hosted by the third-party computing system.

20. The system of claim 15, wherein the operations further comprise: prompting, by the intelligent assistant, the user to upload a document, and wherein the learned attributes of the user is further based on analysis of the uploaded document.
